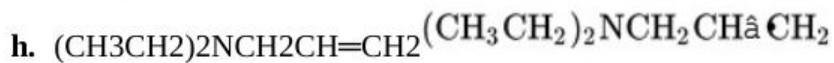
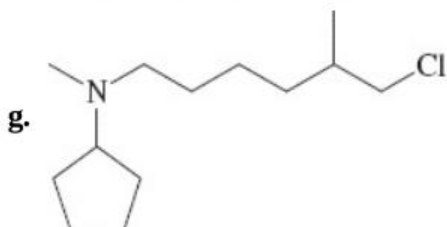
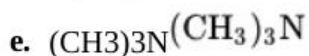
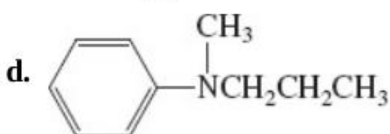
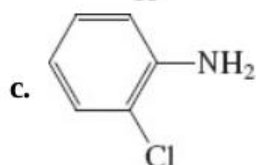
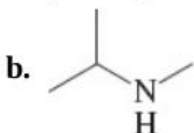
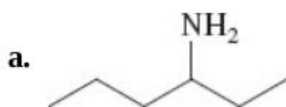


Problems

27. Give at least two names for each of the following amines.



28. Give structures that correspond to each of the following names.

- N,N*-Dimethyl-3-cyclohexenamine;
- N*-ethyl-2-phenylethylamine;
- 2-aminoethanol;
- m*-chloroaniline.

29. Name each of the following compounds pictured in [Real Life 21-1](#) according to IUPAC or *Chemical Abstracts*. Pay attention to the order of

precedence of the functional groups. **(a)** Propylhexedrine; **(b)** amphetamine; **(c)** mescaline; **(d)** epinephrine.

30. As mentioned in [Section 21-2](#), the inversion of nitrogen requires a change of hybridization. **(a)** What is the approximate energy difference between pyramidal nitrogen (sp^3 hybridized) and trigonal planar nitrogen (sp^2 hybridized) in ammonia and simple amines? (**Hint:** Refer to the E_a of inversion.) **(b)** Compare the nitrogen atom in ammonia with the carbon atom in each of the following species: methyl cation, methyl radical, and methyl anion. Contrast the most stable geometries and the hybridizations of each of these species. Using fundamental notions of orbital energies and bond strengths, explain the similarities and differences among them.
31. Use the following NMR- and mass-spectral data to identify the structures of two unknown compounds, A and B.

A: 1H NMR: $\delta = 0.92$ (t, $J = 6$ Hz, 3 H), 1.32 (broad s, 12 H), 1.32 (broad s, 12 H), 2.28 (broad s, 2 H), 2.28 (broad s, 2 H), 2.69 (t, $J = 7$ Hz, 2 H) ppm; ^{13}C NMR: $\delta = 14.1$, six signals between 22.8 and 34.0, 22.8 and 34.0, and 42.4 ppm; MS: m/z (relative intensity) = 129 (0.6), 30 (100).

B: 1H NMR: $\delta = 1.00$ (s, 9 H), 1.17 (s, 6 H), 1.17 (s, 6 H), 1.28 (s, 2 H), 1.28 (s, 2 H), 1.42 (s, 2 H) ppm; ^{13}C NMR: $\delta = 31.7$, 31.7, 32.9, 32.9, 51.2, 51.2, 57.2 ppm; MS: m/z (relative intensity) = 129 (0.05), 114 (3), 114 (3), 72 (4), 72 (4), 58 (100).

32. The following spectroscopic data (^{13}C NMR and IR) are for several isomeric amines of the formula $C_6H_{15}N$. Propose a

structure for each. **(a)** ^{13}C NMR (DEPT): $\delta = 23.7(\text{CH}_3)$,
 ^{13}C NMR (DEPT): $\delta = 23.7(\text{CH}_3)$, 45.3(CH) ppm; 45.3(CH) ppm;
 IR: $\tilde{\nu} = 3300 \text{ cm}^{-1}$ IR: $\tilde{\nu} = 3300 \text{ cm}^{-1}$ **(b)**
 ^{13}C NMR (DEPT): $\delta = 12.6(\text{CH}_3)$,
 ^{13}C NMR (DEPT): $\delta = 12.6(\text{CH}_3)$, 46.9(CH₂) ppm; 46.9(CH₂) ppm;
 IR: no bands in 3250-3500 cm^{-1} range. **(c)**
 ^{13}C NMR (DEPT): $\delta = 12.0(\text{CH}_3)$,
 ^{13}C NMR (DEPT): $\delta = 12.0(\text{CH}_3)$, 23.9(CH₂),
 52.3(CH₂) ppm; 52.3(CH₂) ppm; IR: $\tilde{\nu} = 3280 \text{ cm}^{-1}$
 IR: $\tilde{\nu} = 3280 \text{ cm}^{-1}$. **(d)** ^{13}C NMR (DEPT): $\delta = 14.2(\text{CH}_3)$,
 ^{13}C NMR (DEPT): $\delta = 14.2(\text{CH}_3)$, 23.2(CH₂), 23.2(CH₂), 27.1(CH₂),
 27.1(CH₂), 32.3(CH₂), 32.3(CH₂), 34.6(CH₂), 34.6(CH₂),
 42.7(CH₂) ppm; 42.7(CH₂) ppm; IR: $\tilde{\nu} = 1600 \text{ cm}^{-1}$ (broad),
 IR: $\tilde{\nu} = 1600$ (broad), 3280, 3280, 3365 cm^{-1} . 3365 cm^{-1} . **(e)**
 ^{13}C NMR (DEPT): $\delta = 25.6(\text{CH}_3)$,
 ^{13}C NMR (DEPT): $\delta = 25.6(\text{CH}_3)$, 38.7(CH₃), 38.7(CH₃),
 53.2(C_{quat}) ppm; 53.2(C_{quat}) ppm; IR: no bands in 3250-3500 cm^{-1}
 3250-3500 cm^{-1} region.

33. The following mass-spectral data are for two of the compounds in [Problem 32](#). Match each mass spectrum with one of them. **(a)**
 m/z (relative intensity) = 101 (8), m/z (relative intensity) = 101 (8),
 86 (11), 86 (11), 72 (79), 72 (79), 58 (10), 58 (10), 44 (40), 44 (40),
 30 (100). 30 (100). **(b)** m/z (relative intensity) = 101 (3),
 m/z (relative intensity) = 101 (3), 86 (30), 86 (30), 58 (14), 58 (14),
 44 (100). 44 (100).
34. Is a molecule whose conjugate acid has a high pK_a value a stronger or a weaker base than a molecule whose conjugate acid has a low pK_a value? Explain using a general equilibrium equation.
35. In which direction would you expect each of the following equilibria to lie?
- a. $\text{NH}_3 + ^-\text{OH} \rightleftharpoons \text{NH}_2^- + \text{H}_2\text{O}$

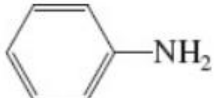


36. (a) How would you expect the following classes of compounds to compare with simple primary amines as bases and acids?

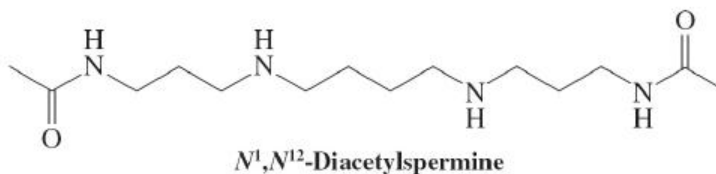
i. Carboxylic amides; for example, CH_3CONH_2 CH_3CONH_2

ii. Imides; for example, $\text{CH}_3\text{CONHCOCH}_3$ $\text{CH}_3\text{CONHCOCH}_3$

iii. Enamines; for example, $\text{CH}_2=\text{CHN}(\text{CH}_3)_2$ $\text{CH}_2=\text{CHN}(\text{CH}_3)_2$

iv. Anilines; for example, 

(b) The tetraamine derivative shown below has been found to be strongly associated with several human cancers. It is relatively scarce, making up less than 0.5% of naturally occurring polyamines in human serum. However, its concentration rises by some 50% in the urine of patients suffering from non-small cell lung cancer, which is the leading cause of cancer deaths worldwide. The detection of this substance holds promise in permitting early-stage diagnosis, when treatment is likely to be more effective. Recent studies showing elevated levels of this substance in colon cancer patients have revealed that it is not only a cancer biomarker, but also a cancer promoter. Thus, bacterial biofilms in the colon favor accumulation of this compound, and antibiotic treatment to clear the biofilms reduces its concentration to normal, thus paving the way for new approaches to control cancer growth.



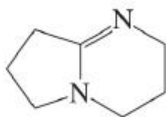
At normal physiological pH this substance exists in a doubly protonated form with pK_a values of 10.2 and 10.6. Write the structures that should be present at $\text{pH} = 7$, 10.4, and 12.

37. Several functional groups containing nitrogen are considerably stronger

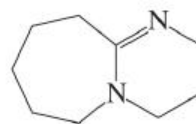
bases than are ordinary amines. One is the amidine group found in DBN and DBU, both of which are widely used as bases in a variety of organic reactions.



Amidine
group



1,5-Diazabicyclo[4.3.0]non-5-ene
(DBN)

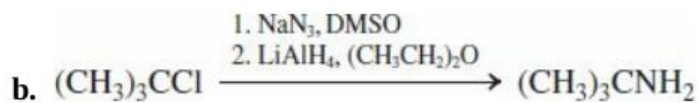
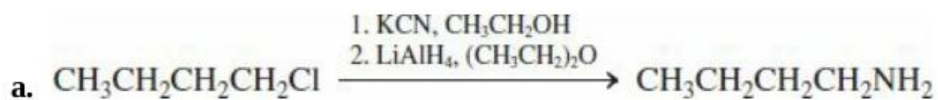


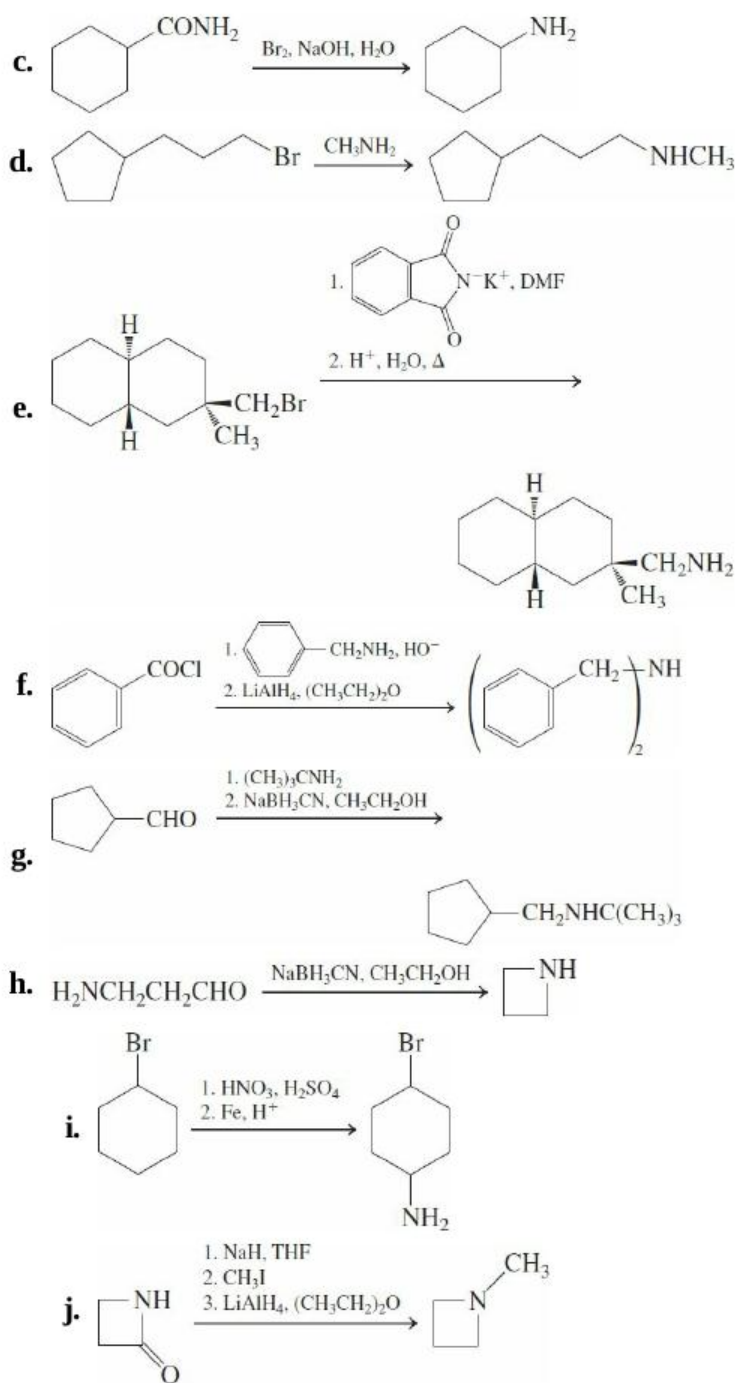
1,8-Diazabicyclo[5.4.0]undec-7-ene
(DBU)



Another unusually strong organic base is guanidine, H_2NCNH_2 . Indicate which nitrogen in each of these bases is the one most likely to be protonated and explain its enhanced base strength relative to that in simple amines.

38. Reaction review: Without consulting the Reaction Road Map on [p. 1058](#), suggest reagents that convert each of the following starting materials into the indicated product: **(a)** (chloromethyl)-benzene (benzyl chloride) into phenylmethanamine (benzylamine); **(b)** benzaldehyde into phenylmethanamine (benzylamine); **(c)** benzaldehyde into *N*-ethylphenylmethanamine (benzylethylamine); **(d)** (chloromethyl)benzene (benzyl chloride) into 2-phenylethanamine (phenethylamine); **(e)** benzaldehyde into *N,N*-dimethylphenylmethanamine (benzyl dimethylamine); **(f)** 1-phenylethanone (acetophenone) into 3-amino-1-phenyl-1-propanone; **(g)** benzonitrile into phenylmethanamine (benzylamine); **(h)** 2-phenylacetamide into phenylmethanamine (benzylamine).
39. In the following syntheses of amines, indicate in each case whether the approach will work well, poorly, or not at all. If a method will not work well, explain why.

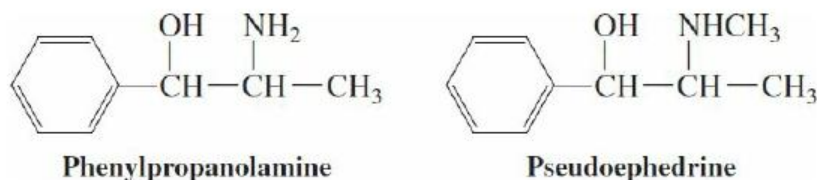




40. For each synthesis in [Problem 39](#) that does not work well, propose an alternative preparation of the final amine, starting either with the same material or with a material of similar structure and functionality.
41. Give the structures of all possible nitrogen-containing organic products

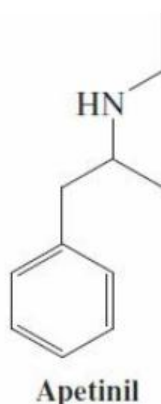
that might be expected to form on reaction of chloroethane with ammonia. (**Hint:** Consider multiple alkylations.)

42. Phenylpropanolamine (PPA) has long been an ingredient in many cold remedies and appetite suppressants. In late 2000, the FDA asked manufacturers to remove products containing this compound from the market because of evidence of an increased risk of hemorrhagic stroke. This action has caused a switch to the safer pseudoephedrine as the active component of such medications.

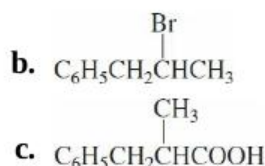


Suppose that you are the director of a major pharmaceutical laboratory with a huge stock of phenylpropanolamine on hand and the president of the company issues the order, “Pseudoephedrine from now on!” Analyze all your options, and propose the best solution that you can find for the problem.

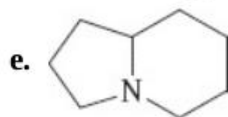
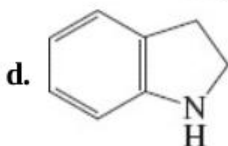
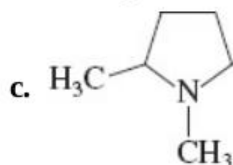
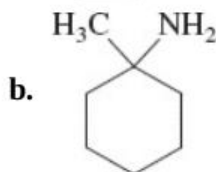
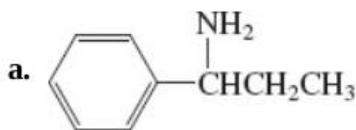
43. Apetinil, an appetite suppressant (i.e., diet pill; see [Real Life 21-1](#)), has the structure shown on the next page. Is it a primary, a secondary, or a tertiary amine? Propose an efficient synthesis of Apetinil from each of the following starting materials. Try to use a variety of methods.



- a. $\text{C}_6\text{H}_5\text{CH}_2\text{COCH}_3$ $\text{C}_6\text{H}_5\text{CH}_2\text{COCH}_3$

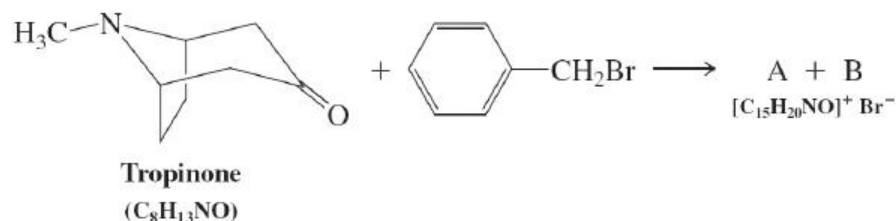


44. Suggest the best syntheses that you can for the following amines, beginning each with any organic starting materials that do not contain nitrogen. **(a)** Butanamine; **(b)** *N*-methylbutanamine; **(c)** *N,N*-dimethylbutanamine.
45. Give the structures of the possible alkene products of Hofmann elimination of each of the following amines. If multiple elimination cycles are possible, give the products of each cycle.



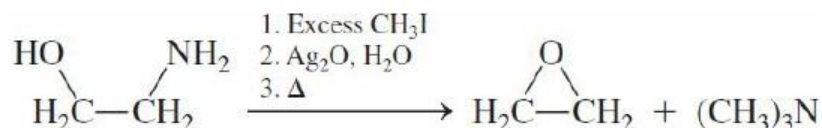
46. What primary amine(s) would give each of the following alkenes or alkene mixtures upon Hofmann elimination? **(a)** 3-Heptene; **(b)** mixture of 2- and 3-heptene; **(c)** 1-heptene; **(d)** mixture of 1- and 2-heptene.
47. Formulate a detailed mechanism for the Mannich reaction between 2-methylpropanal, formaldehyde, and methanamine shown on [p. 1042](#).
48. Reaction of the tertiary amine tropinone with (bromomethyl)- benzene

(benzyl bromide) gives not one but two quaternary ammonium salts, A and B.

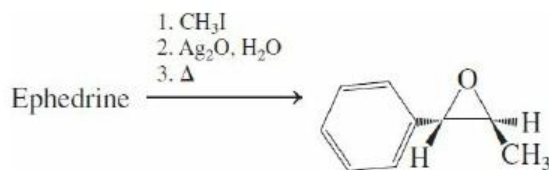


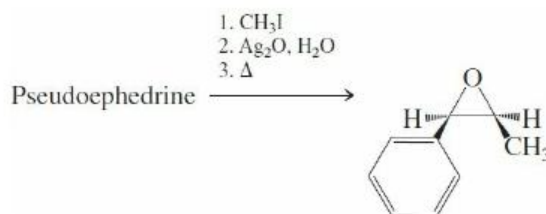
Compounds A and B are stereoisomers that are interconverted by base; that is, base treatment of either pure isomer leads to an equilibrium mixture of the two. **(a)** Propose structures for A and B. **(b)** What kind of stereoisomers are A and B? **(c)** Suggest a mechanism for the equilibration of A and B by base. (**Hint:** Think “reversible Hofmann elimination.”)

49. Attempted Hofmann elimination of an amine containing a hydroxy group on the β -carbon gives an oxacyclopropane product instead of an alkene.

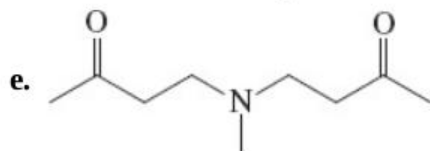
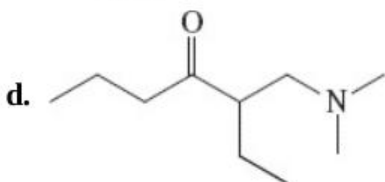
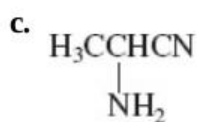
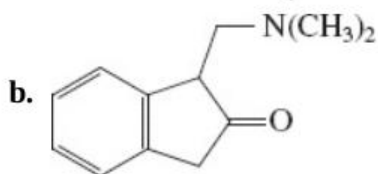
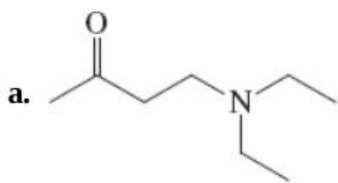


(a) Propose a sensible mechanism for this transformation. **(b)** Pseudoephedrine (see [Problem 42](#)) and ephedrine are closely related natural products, as the similar names imply. In fact, they are stereoisomers. From the results of the following reactions, deduce the precise stereochemistries of ephedrine and pseudoephedrine.





50. Show how each of the following molecules might be synthesized by Mannich or Mannich-like reactions. (**Hint:** Work backward, identifying the bond made in the Mannich reaction.)



51. Tropinone ([Problem 48](#)) was first synthesized by Sir Robert Robinson (famous for the Robinson annulation reaction; [Section 18-11](#)), in 1917, by the following reaction. Show a mechanism for this transformation.